

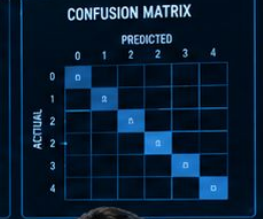
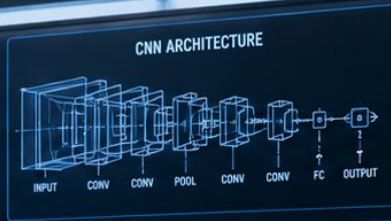
# ★ THE INTERNATIONAL MUSTACHE AUTHORITY

★ Facial Hair Research Division ★

TensorFlow • CNN Neural Networks

Emil Sjökvist • Nicklas Thegerström • Fredrik Käll

I.M.A.  
INTegrity.  
METRICS.  
MUSTACHES.





# CELEBA DATASET

CELEBFACES ATTRIBUTES DATASET



**202,599**  
IMAGES



**40**  
FACIAL ATTRIBUTES

DATASET STATUS  
**VERIFIED**



INTERNATIONAL MUSTACHE AUTHORITY



RESEARCH DIVISION



ATTRIBUTE CATEGORIES



DISTRIBUTION OVERVIEW



SOURCE

CELEBA  
v1.0







# MODEL 1: MUSTACHE DETECTION

FIRST CLASSIFIER: MUSTACHE OR CLEAN

## DATASET CREATION

SOURCE: CELEBA (MALE ONLY)

```
mustache_df = men[men['Mustache'] == 1]
clean_df     = men[men['Mustache'] == -1]
```

```
mustache_sample = mustache_df.sample(
    n=4000, random_state=42)
clean_sample     = clean_df.sample(
    n=4000, random_state=42)
```

Total män: {len(men);}

MUSTACHE: 4,000

CLEAN: 4,000

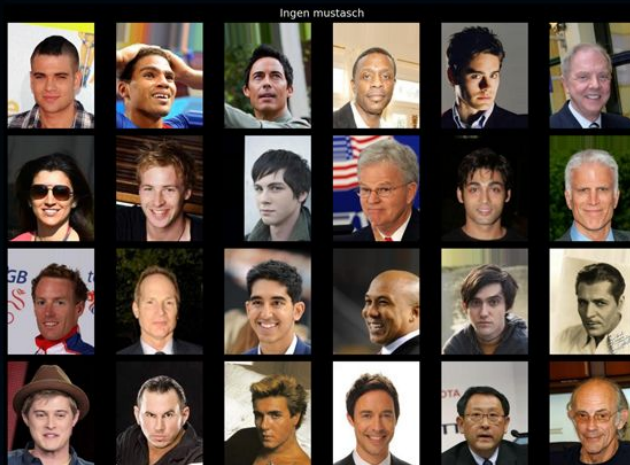
## DATA LOADING (TENSORFLOW)

```
train_ds = tf.keras.utils.image_dataset_from_directory(
    'data/dataset_v2',
    validation_split=0.2,
    subset='training',
    seed=42,
    image_size=(128, 128),
    batch_size=32,
    label_mode='binary')
```

```
val_ds = tf.keras.utils.image_dataset_from_directory(
    'data/dataset_v2',
    validation_split=0.2,
    subset='validation',
    seed=42,
    image_size=(128, 128),
    batch_size=32,
    label_mode='binary')
```

Klasser: ['clean', 'mustache']

## CLEAN (INGEN MUSTASCH)



## MUSTACHE



8,000  
TOTAL IMAGES



4,000  
CLEAN

4,000  
MUSTACHE



80%  
TRAINING  
20%  
VALIDATION



BINARY  
CLASSIFICATION  
OUTPUT: 2 CLASSES



# MODEL 1: EVALUATION

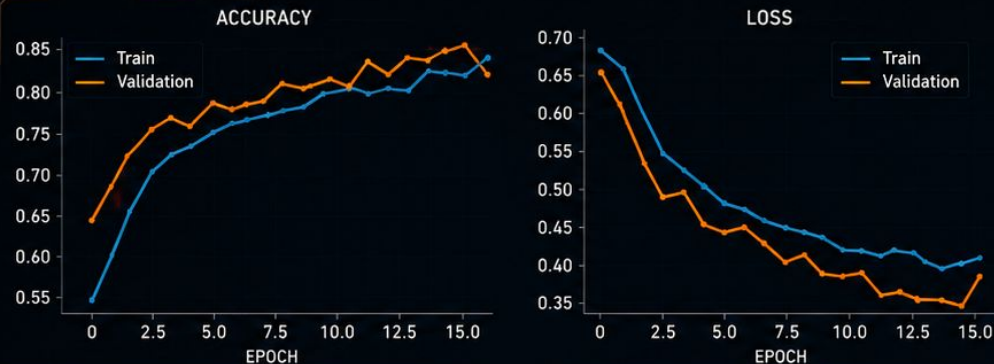
## MUSTACHE DETECTOR CNN



IMA-17A

CLEARANCE LEVEL: GOLD

### TRAINING HISTORY



### CONFUSION MATRIX



# 83.9%

VALIDATION ACCURACY



# 15

EPOCHS



## BINARY CLASSIFICATION

OUTPUT: 2 CLASSES



# 622

CORRECT  
CLEAN PREDICTIONS

# 721

CORRECT  
MUSTACHE PREDICTIONS

# 165

FALSE  
POSITIVES

# 92

FALSE  
NEGATIVES

### TRAINING CONFIGURATION

```
model_v2.compile(  
    optimizer='adam',  
    loss='binary_crossentropy',  
    metrics=['accuracy']  
)  
early_stop = tf.keras.callbacks.EarlyStopping(  
    monitor='val_loss',  
    patience=5,  
    restore_best_weights=True,  
    verbose=1  
)
```

```
history = model_v2.fit(  
    train_ds,  
    validation_data=val_ds,  
    epochs=15,  
    callbacks=[early_stop]  
)
```







★ RESEARCH DIVISION ★

— CLASSIFIED RESEARCH MATERIAL —

# MODEL 1.5: DATASET SANITIZATION

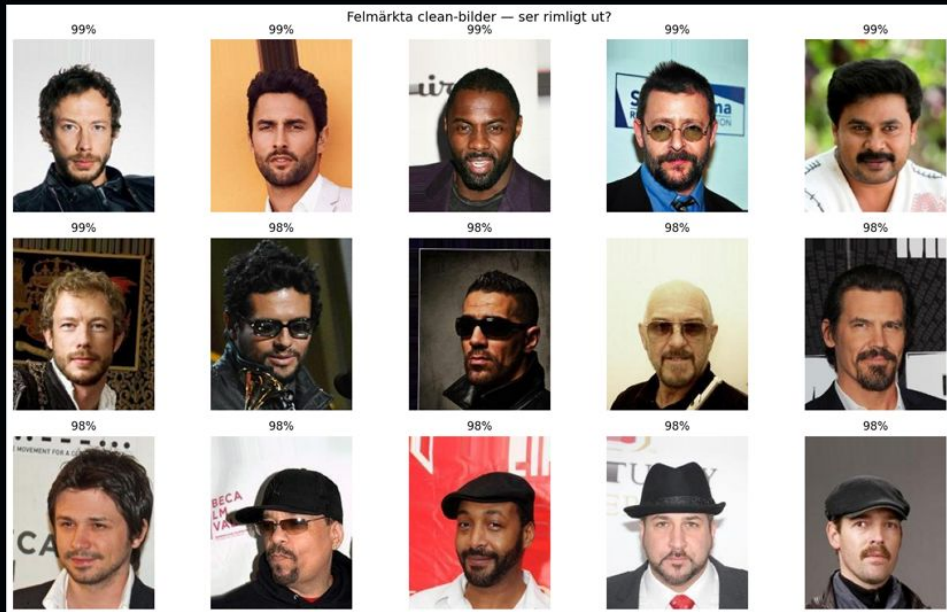
USING AI TO CLEAN THE TRAINING DATA



IMA-17A

CLEARANCE LEVEL: GOLD

## FALSE "CLEAN" EXAMPLES DISCOVERED BY MODEL 1



## CONFUSION MATRIX (VALIDATION SET)



## DATASET CLEANING SUMMARY



## MODEL PERFORMANCE



IMA-17A  
CLEARANCE LEVEL: GOLD

★ THE MUSTACHES WERE HIDING IN PLAIN SIGHT. ★

— EXPERIMENT STATUS: DATASET SANITIZED —

RESEARCH LOG  
DATE: 2024-05-22





★ RESEARCH DIVISION ★

★ CLASSIFIED RESEARCH MATERIAL ★

# MANUAL EPICNESS CLASSIFICATION

*The Burden of Science*

EVERY MUSTACHE  
TELLS A STORY.  
ONLY THE EPIC ONES  
MAKE HISTORY.



# 4,414

MUSTACHES WERE EXAMINED

# 597

WERE DEEMED WORTHY



## NEW EPIC DATASET

EPIC  
**334**

THIN  
**263**

TOTAL IMAGES  
**597**

### EPICNESS EVALUATION FORM

- ☒ THICKNESS
- ☒ DENSITY
- ☒ SHAPE
- ☒ SYMMETRY
- ☒ OVERALL IMPACT



REJECTED  
4,414

MUSTACHE  
REPORTS

INTERNATIONAL  
MUSTACHE AUTHORITY

IMA-17A

CLEARANCE LEVEL: GOLD

★ EVERY IMAGE WAS MANUALLY REVIEWED BY THE INTERNATIONAL MUSTACHE AUTHORITY ★

RESEARCH LOG

DATE: 2024-05-22





• RESEARCH DIVISION •

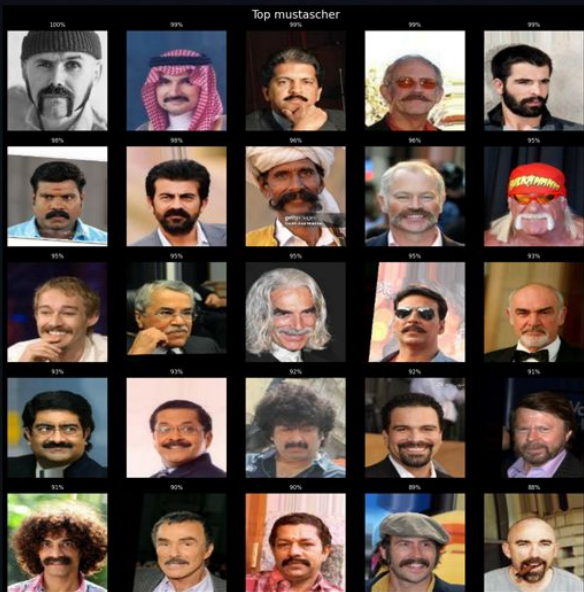
• CLASSIFIED RESEARCH MATERIAL

# MODEL 2: EPIC MUSTACHE CLASSIFIER

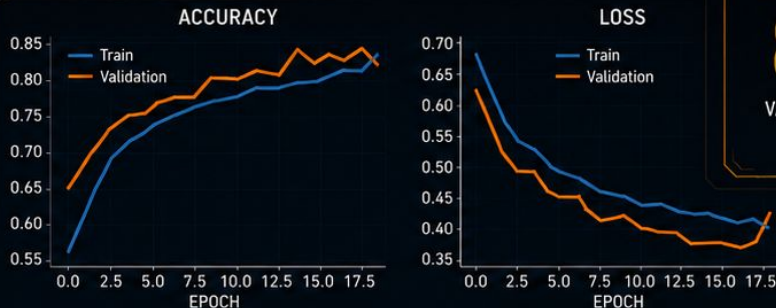
EPIC DETECTOR — MOBILENETV2

IMA-17A  
CLEARANCE LEVEL: GOLD

## TOP 25 MOST EPIC MUSTACHES



## TRAINING HISTORY



## VALIDATION ACCURACY

**86.6%**

VALIDATION LOSS: 0.3694



## CONFUSION MATRIX (VALIDATION SET)

|            |      | PREDICTED LABEL |      |
|------------|------|-----------------|------|
|            |      | EPIC            | THIN |
| TRUE LABEL | EPIC | 57              | 9    |
|            | THIN | 7               | 46   |

## NEW EPIC DATASET

 EPIC  
**334** THIN  
**263**

TOTAL IMAGES

**597**BUILT ON MANUALLY CURATED DATA  
EVERY IMAGE REVIEWED BY EXPERTSBINARY CLASSIFICATION  
EPIC = 1 | THIN = 0MISSION: IDENTIFY TRUE EPICNESS  
ONE MUSTACHE AT A TIME

SCIENCE. DISCIPLINE. DIGNITY. MUSTACHES.



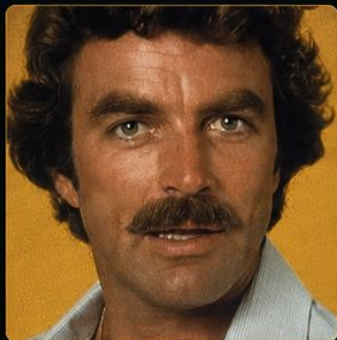
[moustacheclassifier.streamlit.app](https://moustacheclassifier.streamlit.app)



## SUBMITTED SPECIMEN

Official Facial Hair Assessment by the International Mustache Authority

Submit specimen for analysis



Submitted specimen

Initiate Analysis

## ANALYSIS RESULTS

mustache\_prob: 0.9140045046806335

thin\_prob: 0.0927349105477333

epic\_score: 0.9979915983974934

## MUSTACHE POWER RATING™

99.8 / 100



## LEGENDARY STACHE

The International Mustache Authority is impressed.

Results certified by the International Mustache Authority™



IMA-17A

CLEARANCE LEVEL: GOLD



UPLOAD



ANALYZE



RESULTS



REPORT

RESEARCH LOG

DATE: 2024-05-22

SCIENCE.  
DISCIPLINE.  
DIGNITY.  
MUSTACHES.

IMA-17A

EVERY MUSTACHE  
TELLS A STORY.  
ONLY THE EPIC ONES  
MAKE HISTORY.INTERNATIONAL  
MUSTACHE AUTHORITY

TOP SECRET

SPECIMEN ANALYSIS REPORT

EPIC



# PROBLEMS & SOLUTIONS

EVERY MUSTACHE  
TELLS A STORY.  
ONLY THE EPIC ONES  
MAKE HISTORY.



| PROBLEM                                                                                                                                                       | SOLUTION                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Label noise in CelebA — mustache images incorrectly labeled as clean</p> |  <p>Automatic dataset cleaning using the first version of the model</p>                                   |
|  <p>The model struggled with zoomed-out faces</p>                            |  <p>MTCNN face cropping to normalize facial input</p>                                                     |
|  <p>The clean class contained only clean-shaven men</p>                      |  <p>Added 1,382 women to the training dataset, replacing mislabeled clean samples detected by Model 1</p> |

